

1. For an ideal gas,

$$P = \frac{1}{3}\rho\overline{v^2},$$

where ρ is the density of the gas.

- (a) Find the root-mean-square speed of H_2 molecules at STP.

Given: $\rho_{STP} = 0.09 \text{ kg/m}^3$, $P = 760 \text{ mmHg} = 101325 \text{ Pa}$, $T_{STP} = 273.15 \text{ K}$,

Find: v_{rms} .

$$\begin{aligned} P &= \frac{1}{3}\rho\overline{v^2} \\ \overline{v^2} &= 3\frac{P}{\rho} \\ \sqrt{\overline{v^2}} &= \sqrt{3\frac{P}{\rho}} \\ v_{rms} &= \sqrt{3\frac{P}{\rho}} \\ &= \sqrt{3\frac{101325 \text{ Pa}}{0.09 \text{ kg/m}^3}} \\ &= 1837.8 \text{ m/s.} \end{aligned}$$

$\approx 1840 \text{ m/s}$

- (b) Is the answer in part (a) reasonable?

Yes, the answer in part (a) is reasonable. The root-mean-square speed of H_2 molecules at STP is known to be around 1838 m/s, and the answer that I reached is close to this speed.

2. Derive the barometric equation.

$$\begin{aligned} F_{above} + mg &= F_{below} \\ F_{above} - F_{below} &= -mg \\ (P_{above} - P_{below})A &= -mg \\ dPA &= -mg \\ dP &= -\frac{mg}{A} \\ &= -\frac{\rho Vg}{A} \\ &= -\rho g dz \\ \frac{dP}{dz} &= -\rho g. \\ PV &= Nk_B T \end{aligned}$$

$$P = \frac{N}{V}k_B T$$

$$\rho = m \frac{N}{V}$$

$$P = \frac{\rho k_B T}{m}$$

$$\rho = \frac{m}{k_B T} P$$

$$\frac{dP}{dz} = -\frac{mg}{k_B T} P.$$